

Switching

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Switching

Whenever we have multiple devices: how to connect??

- make a point-to-point connection between each pair of devices (a mesh topology) ?
- make a connection between a central device and every other device (a star topology)?

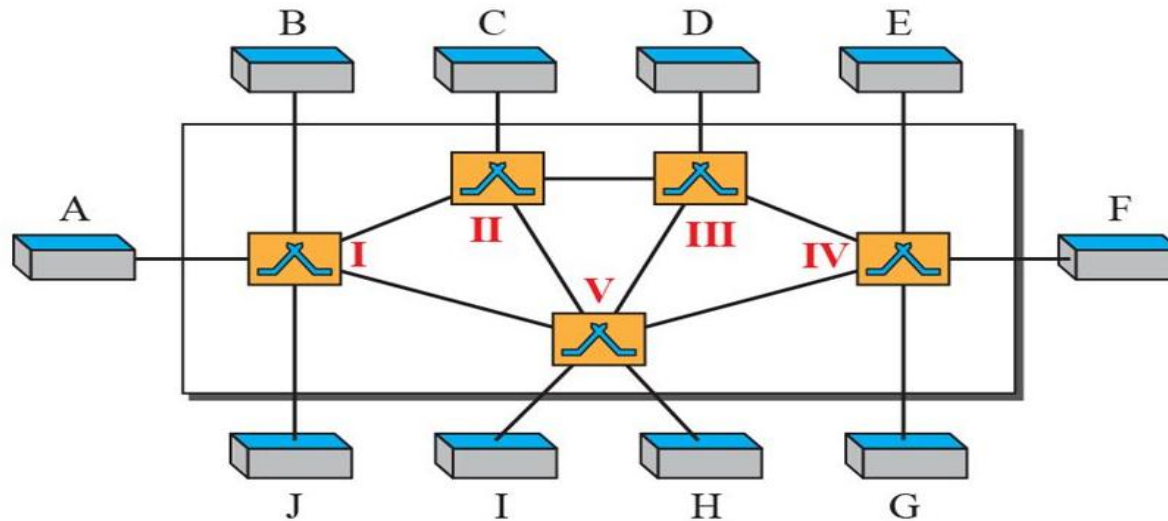
impractical and wasteful when applied to very large networks.

- The number and length of the links require too much infrastructure to be cost-efficient
- the majority of those links would be idle most of the time.

A better solution is switching.

A switched network consists of a series of interlinked nodes, called switches creating temporary connections between two or more devices linked to the switch.

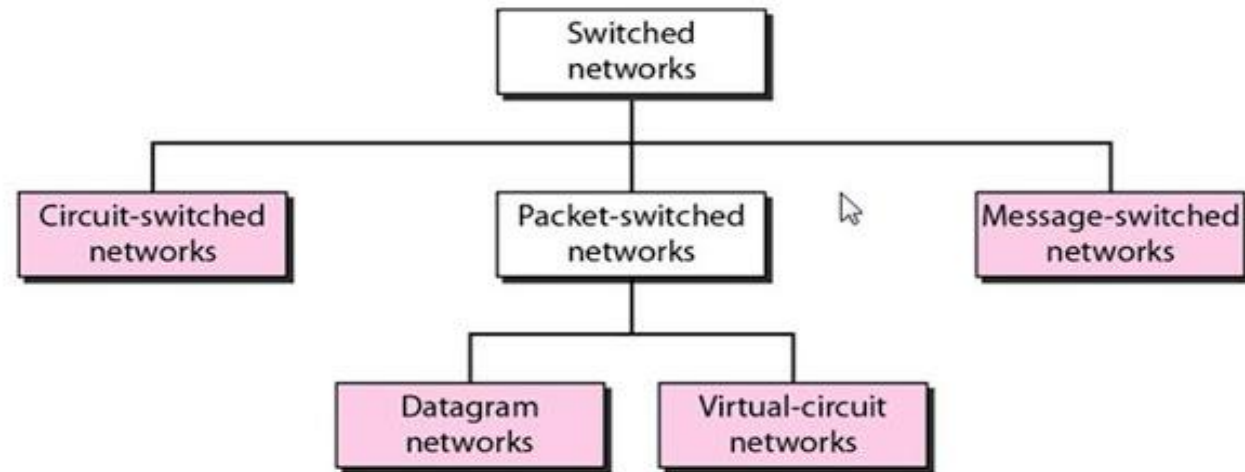
Switched Network



The end systems (communicating devices) are labeled A, B, C, D, and so on, switches are labeled I, II, III, IV, and V

Switching is the technique used in computer networks to **transfer data from a source to a destination through intermediate nodes (switches/routers)** by selecting an appropriate path.

Switching Methods



Today the tendency in packet switching is to combine datagram networks and virtualcircuit networks.

Networks route the **first packet based on the datagram addressing idea**, but **then create a virtual-circuit network for the rest of the packets** coming from the same source and going to the same destination.

Message Switching

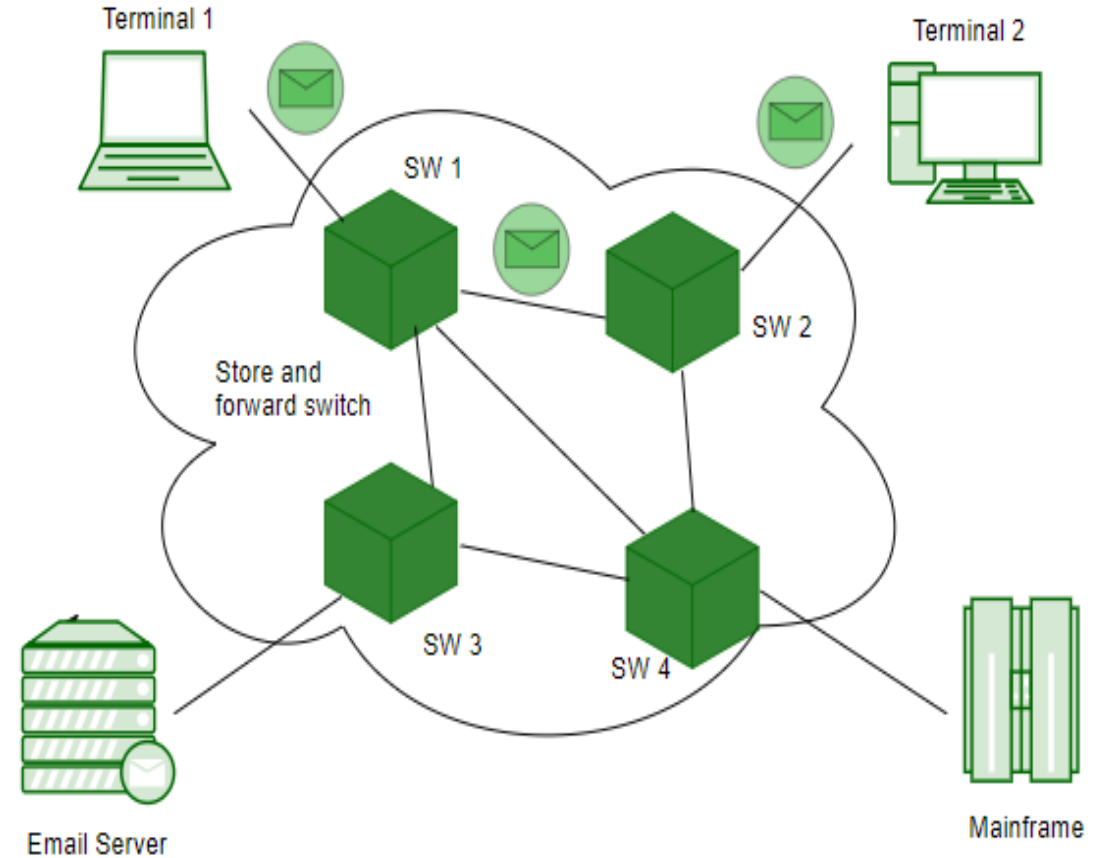
- **Message switching** is a switching technique in which the **entire message** is transmitted from source to destination **node by node** using the **store-and-forward** method.
- There is **no dedicated path** between sender and receiver.
- *Each intermediate node receives the full message, stores it temporarily, and then forwards it to the next node.*

Example

- **Telegraph systems**
- **Early email systems**
- **Old military communication networks**

Working of Message Switching

1. Sender creates the **complete message**
 2. Message is sent to the **first switching node**
 3. The switch:
 1. Stores the full message
 2. Checks destination address
 3. Forwards it when the next link is free
 4. This process repeats at each node
 5. Destination receives the **entire message**
- There is **no setup phase** and **no real-time transmission**.



Key Characteristics

- Store and forward technique
- No dedicated communication path
- Large storage required at switches
- High and unpredictable delay
- No call blocking

Advantages

- No need for a dedicated line
- Efficient link utilization
- No call blocking
- Flexible routing

Disadvantages

- Very high delay
- Large buffer/storage needed
- Not suitable for real-time communication
- Obsolete technology

CIRCUIT-SWITCHING

- **Circuit switching** is a switching technique in which a **dedicated physical communication path** is established between sender and receiver **before** data transmission begins.
- This path remains **reserved exclusively** for the entire duration of communication.

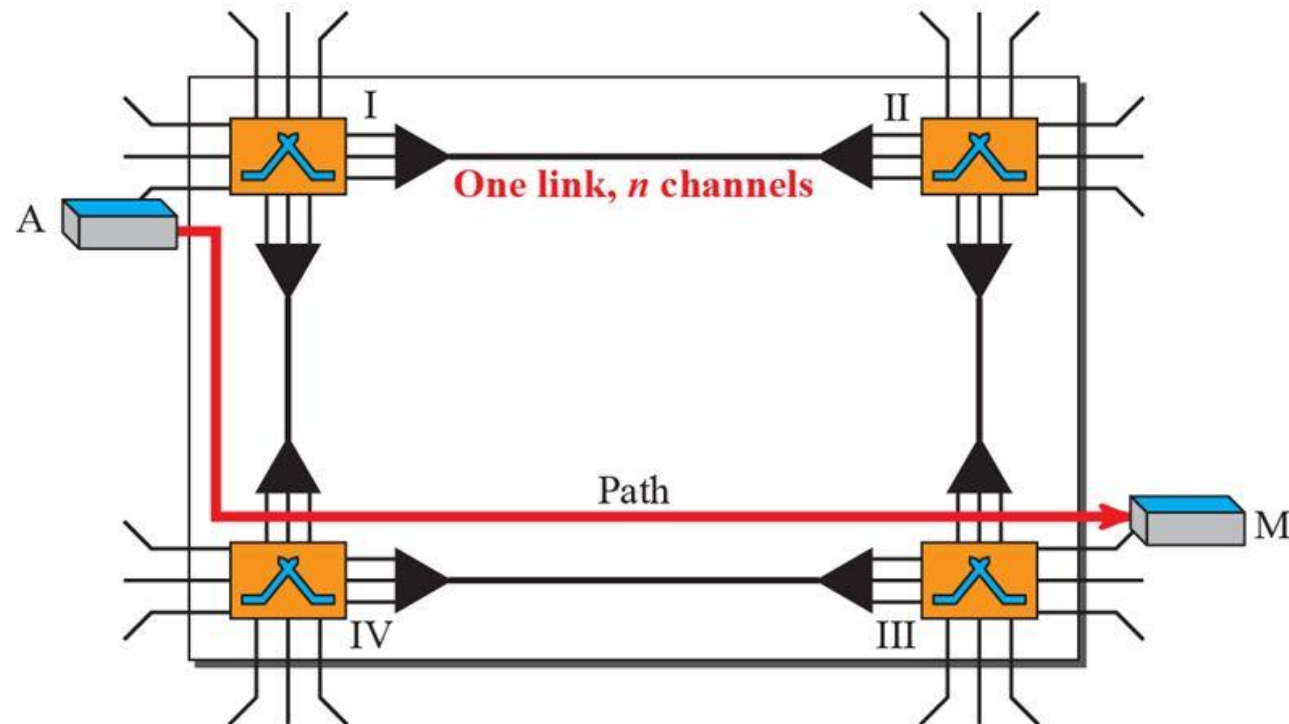
Once the circuit is set up, data flows continuously without interruption.

Example

- **Traditional Telephone Network (PSTN)**
- ISDN
- Leased lines

CIRCUIT-SWITCHED NETWORKS

each connection uses only one **dedicated channel** on each link. Each link is normally divided into n channels by using *FDM or TDM*



CIRCUIT-SWITCHED NETWORKS

- Circuit switching takes place at the **physical layer**
- Before starting communication, **make a reservation** for the resources (**Setup Phase**)
- resources, such as channels (bandwidth in FDM and time slots in TDM), switch buffers, switch processing time, ...
- Data transferred between the two stations **are not packetized** (physical layer transfer of the signal).
- There is **no addressing involved during data transfer**. The switches route the data based on their occupied band (FDM) or time slot (TDM).
- there is **end-to-end addressing used during the setup phase**

Working of Circuit Switching

1. Circuit Establishment (Setup Phase)

- Sender sends a **call request**
- Switching devices (exchanges) find a free path
- Resources like **bandwidth, channels, and switches** are reserved
- Receiver sends **acknowledgement**

No data is transmitted until setup is complete.

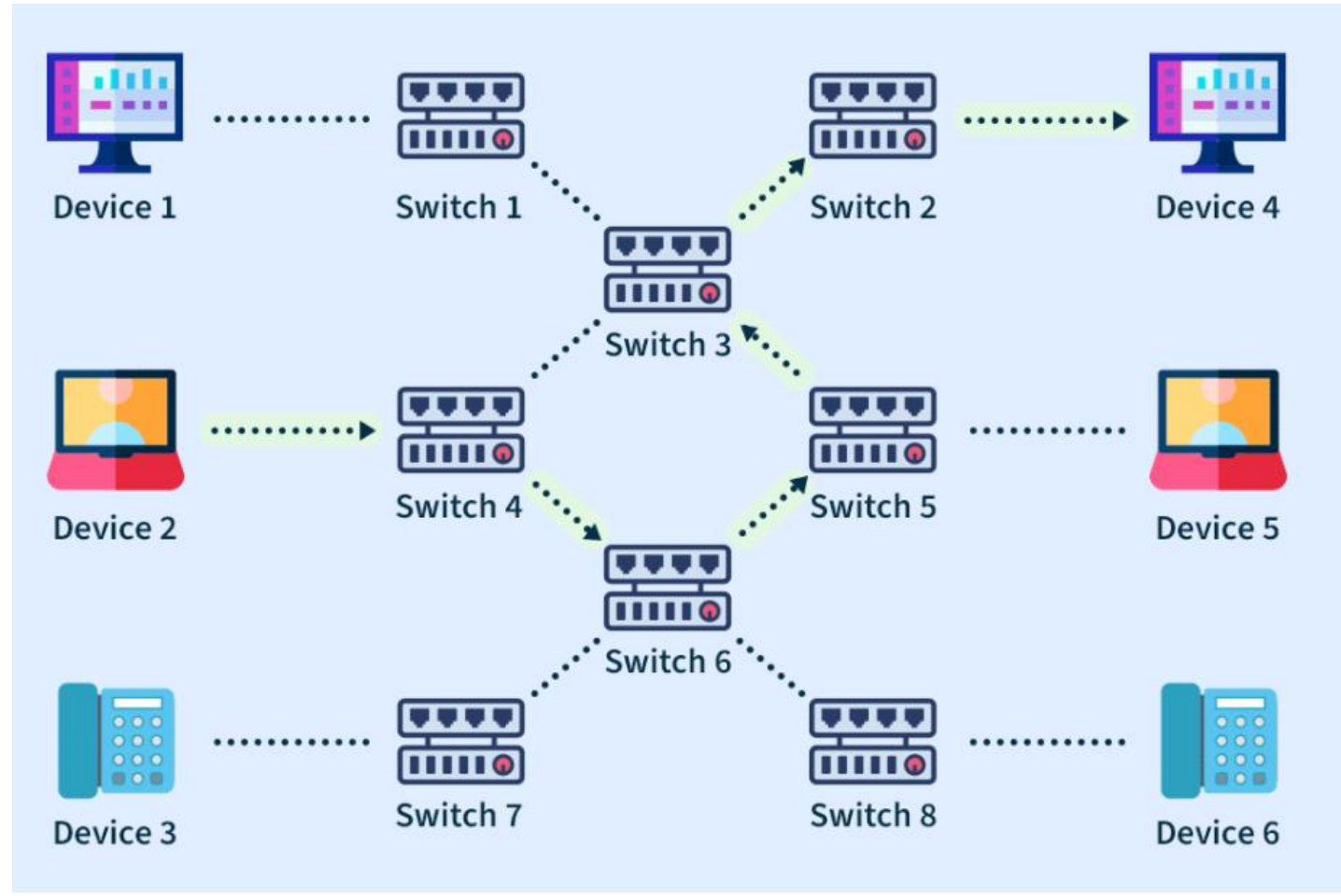
2. Data Transfer Phase

- Data is transmitted in a **continuous stream**
- No addressing information is required in data
- Fixed data rate
- No packet loss or reordering

Communication is **real-time and reliable**.

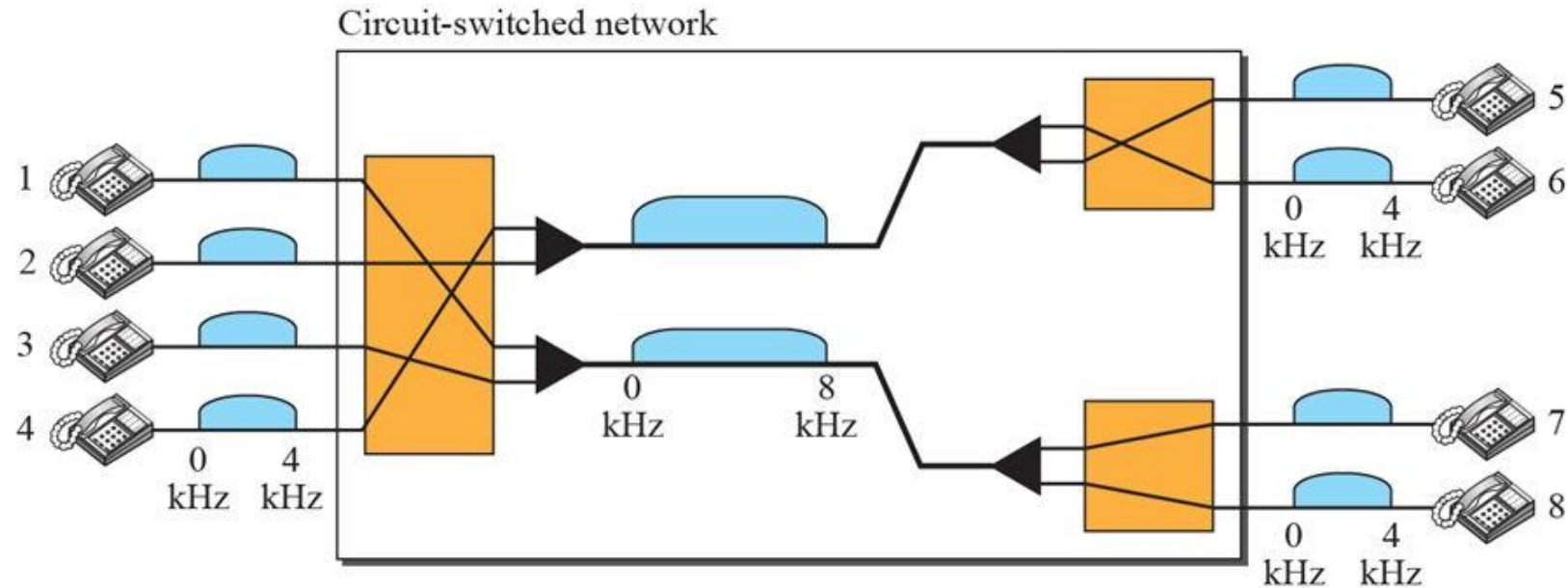
3. Circuit Release (Teardown Phase)

- After communication ends:
 - Circuit is disconnected
 - Resources are freed
- Ready for next call



CIRCUIT-SWITCHED NETWORKS

Example 1



- Telephone 1 is connected to telephone 7; 2 to 5; 3 to 8; and 4 to 6
- The situation may change when new connections are made

CIRCUIT-SWITCHED NETWORKS

Setup Phase

- Before the two parties can communicate, a dedicated circuit needs to be established.
- connection setup means creating dedicated channels between the switches.
- end-to-end addressing is required for creating a connection between two end systems. These can be, for example, the addresses of the computers assigned by the administrator in a TDM network, or telephone numbers in an FDM network.

Data Transfer Phase

- After the establishment of the dedicated circuit (channels), the two parties can transfer data

Teardown Phase

- When one of the parties needs to disconnect, a signal is sent to each switch to release the resources

CIRCUIT-SWITCHED NETWORKS

Efficiency

- are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection and are unavailable to other connections.

Delay

- the delay in this type of network is minimal

CIRCUIT-SWITCHED NETWORKS

Delay

delay caused by the setup is the sum of four parts:

- the **propagation time** of the source computer request (slope of the first gray box),
- the **request signal transfer time** (height of the first gray box),
- the **propagation time of the acknowledgment** from the destination computer (slope of the second gray box),
- the **signal transfer time** of the acknowledgment (height of the second gray box).

The delay due to data transfer is the sum of two parts:

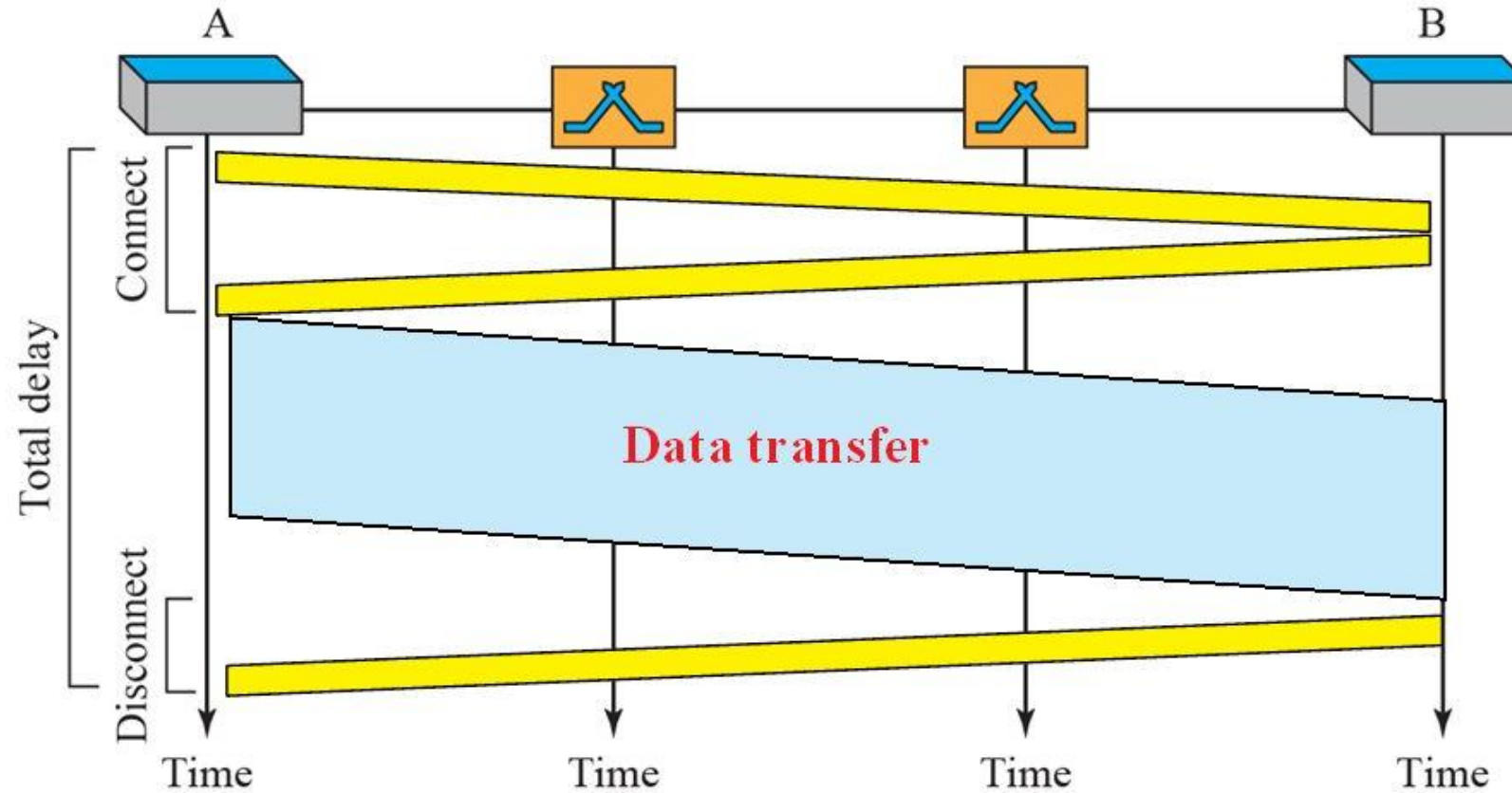
- the propagation time (slope of the colored box)
- data transfer time (height of the colored box), which can be very long.

The third box shows the time needed to tear down the circuit.

- We have shown the case in which the receiver requests disconnection, which creates the maximum delay.

CIRCUIT-SWITCHED NETWORKS

Delay



Example

A path in a digital circuit-switched network has a data rate of 1 Mbps. The exchange of 1000 bits is required for the setup and teardown phases. The distance between two parties is 5000 km. Answer the following questions if the propagation speed is 2×10^8 m:

- a. What is the total delay if 1000 bits of data are exchanged during the data transfer phase?
- b. What is the total delay if 100,000 bits of data are exchanged during the data transfer phase?
- c. What is the total delay if 1,000,000 bits of data are exchanged during the data transfer phase?
- d. Find the delay per 1000 bits of data for each of the above cases and compare them. What can you infer?

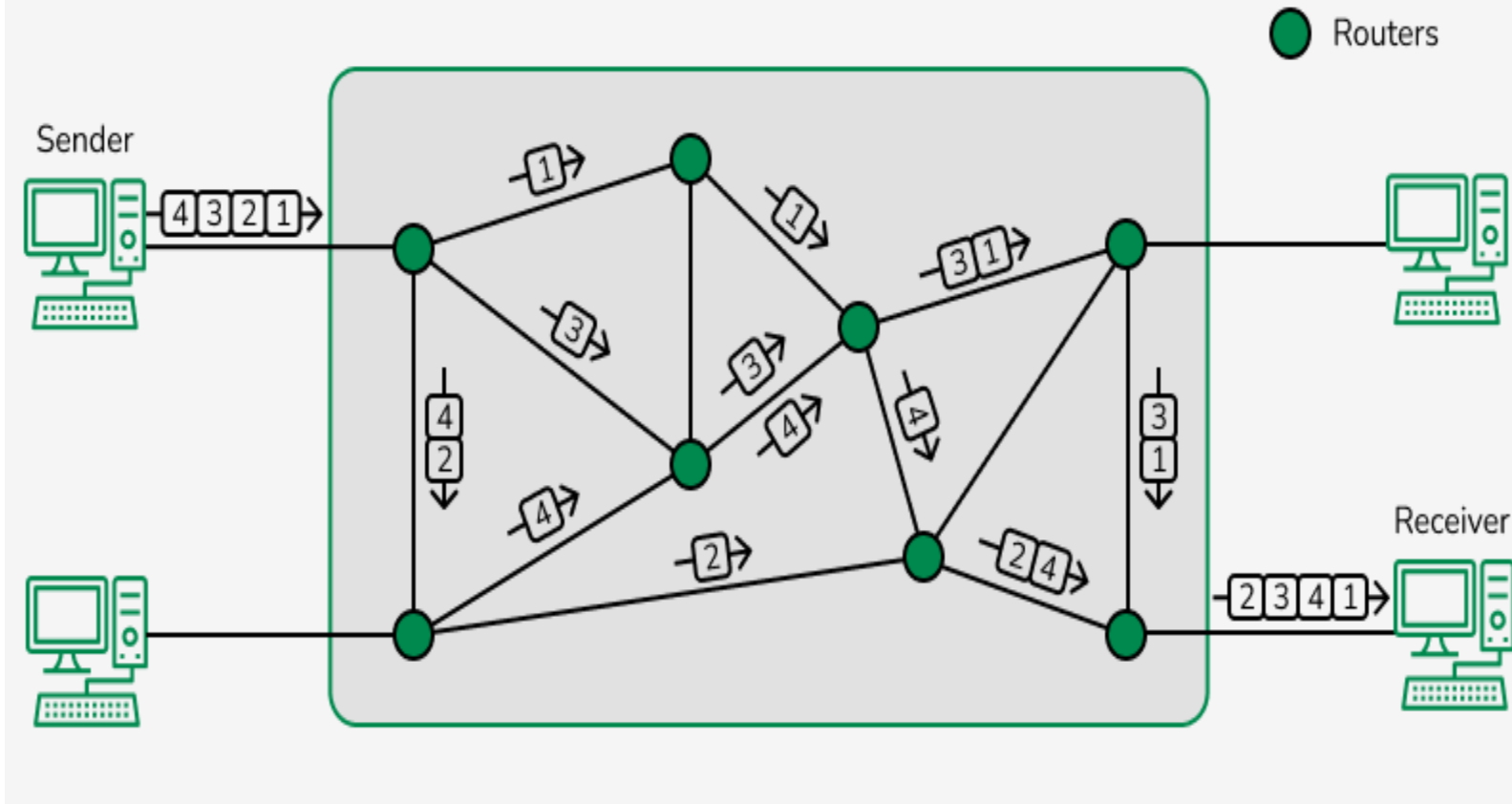
PACKET SWITCHING

- **Packet switching** is a switching technique in which a message is **divided into small packets**, and each packet is transmitted independently over a **shared network**.
- Packets are reassembled at the destination to form the original message.
- *No dedicated path is required.*

How Packet Switching Works

1. Sender divides data into **packets**
2. Each packet contains:
 - Source address
 - Destination address
 - Sequence number
 - Error control information
3. Packets are sent into the network
4. Routers:
 - Choose the best available path
 - Forward packets independently
5. Destination:
 - Receives packets (may be out of order)
 - Reorders and reconstructs data

Packet Switching



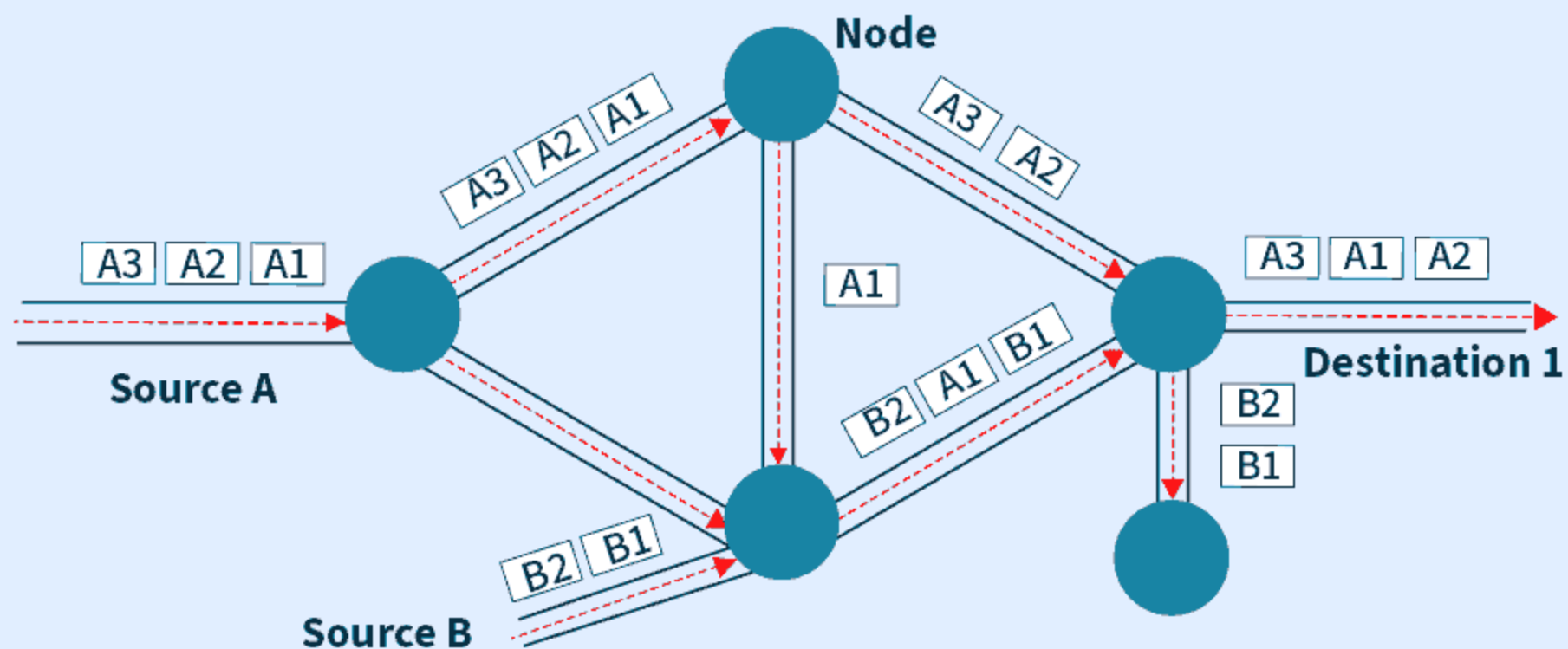
1. Datagram Packet Switching

- In **Datagram Packet Switching**, each packet is treated independently.
- There is **no pre-established path** between sender and receiver.
- Each packet contains the **complete destination address**, and routers decide the **best path for every packet separately**.

DATAGRAM NETWORKS

- messages needs to be divided into packets of fixed or variable size
- there is no resource allocation for a packet and are allocated on demand
- The allocation is done on a first-come, first-served basis
- each packet is treated independently of all others
- Packets in this approach are referred to as datagrams
- Datagram switching is normally done at the network layer

Datagram Switching

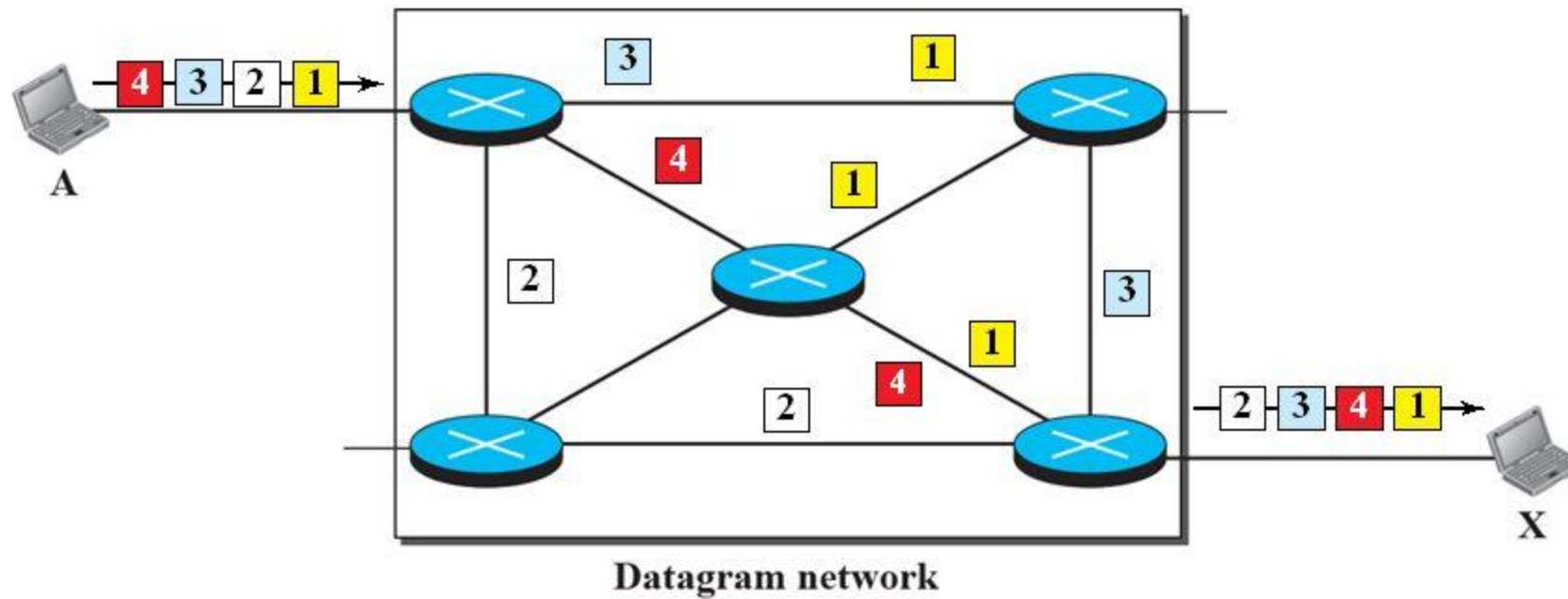


DATAGRAM NETWORKS

may travel **different paths** to reach their destination

- **out of order Transfer**
- **different delays** between the Packets
- Packets may be **lost or dropped**
- In most protocols, it is the responsibility of an **upper-layer protocol** to reorder the datagrams or ask for lost datagrams before passing them on to the application.
- There are **no setup or teardown phases** (**connectionless networks**)

DATAGRAM NETWORKS



Characteristics

- No connection setup phase
- No resource reservation
- Dynamic routing
- Best-effort delivery
- Packets may be lost, delayed, or duplicated

Advantages

- Very efficient use of bandwidth
- Highly flexible
- Fault tolerant (alternate routes possible)
- Suitable for bursty data

Disadvantages

- Packet loss possible
- Delay and jitter
- Packets may arrive out of order
- Not ideal for real-time apps without QoS

Routing Table

- If there are no setup or teardown phases, how are the packets routed to their destinations in a datagram network?
- each switch has a routing table which is based on the destination address
- The destination addresses and the corresponding forwarding output ports are recorded in the tables
- In table of a circuit switched network , each entry is created when the setup phase is completed and deleted when the teardown phase is over.

Routing Table

- Every packet in a datagram network carries destination address of the packet.
- the routing table is consulted to find the corresponding port through which the packet should be forwarded
- The address remains the same during the entire journey of the packet opposite in a virtual-circuit-switched network

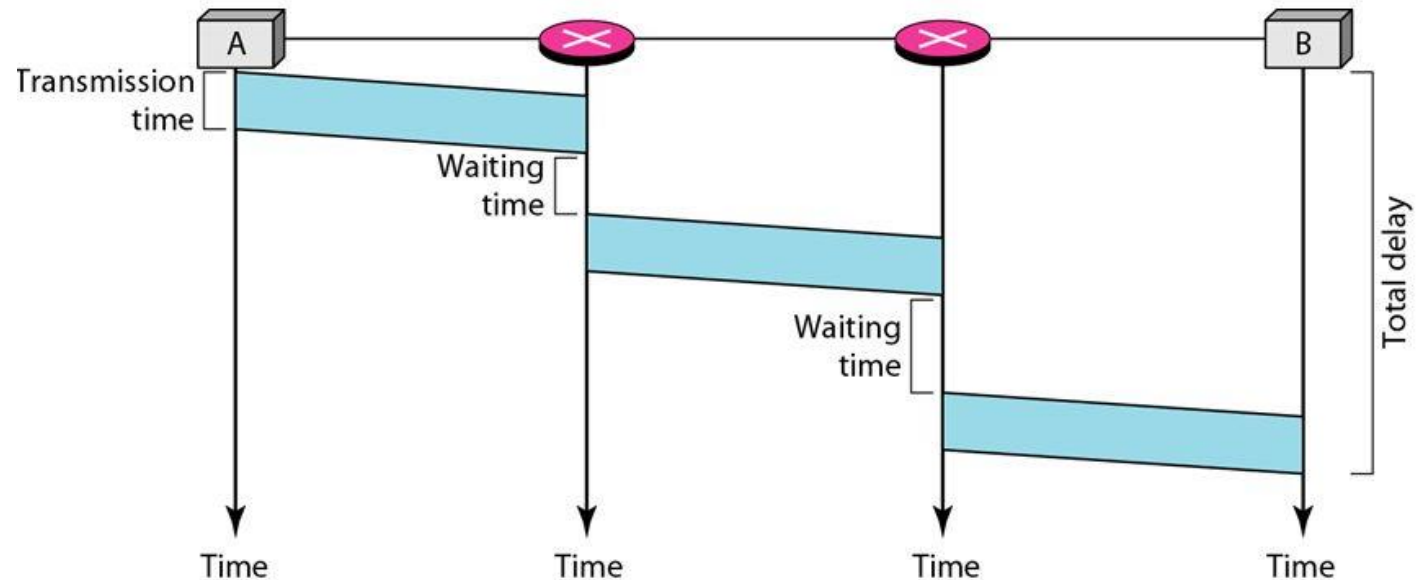
Efficiency

- better than that of a circuit-switched network
- resources are allocated only when there are packets to be transferred

Delay

- May be greater delay in a datagram network than in a virtual-circuit network
- each packet may experience a wait at a switch before it is forwarded
- the delay is not uniform for the packets of a message

Delay



three transmission times ($3T$),
three propagation delays (slopes $3t$ of the lines),
two waiting times ($W1 + W2$)
ignore the processing time in each switch.
The total delay is **Total delay = $3T + 3t + W1 + W2$**

2. Virtual Circuit Packet Switching

- In **Virtual Circuit Packet Switching**, a **logical path (virtual circuit)** is established **before data transmission**, but data is still sent in **packets**.
- All packets follow the **same path**.

Phases of Virtual Circuit Switching

1. Setup Phase

- Route is established
- Virtual Circuit Identifier (VCI) assigned

2. Data Transfer Phase

- Packets follow the **same path**
- Packets arrive **in order**

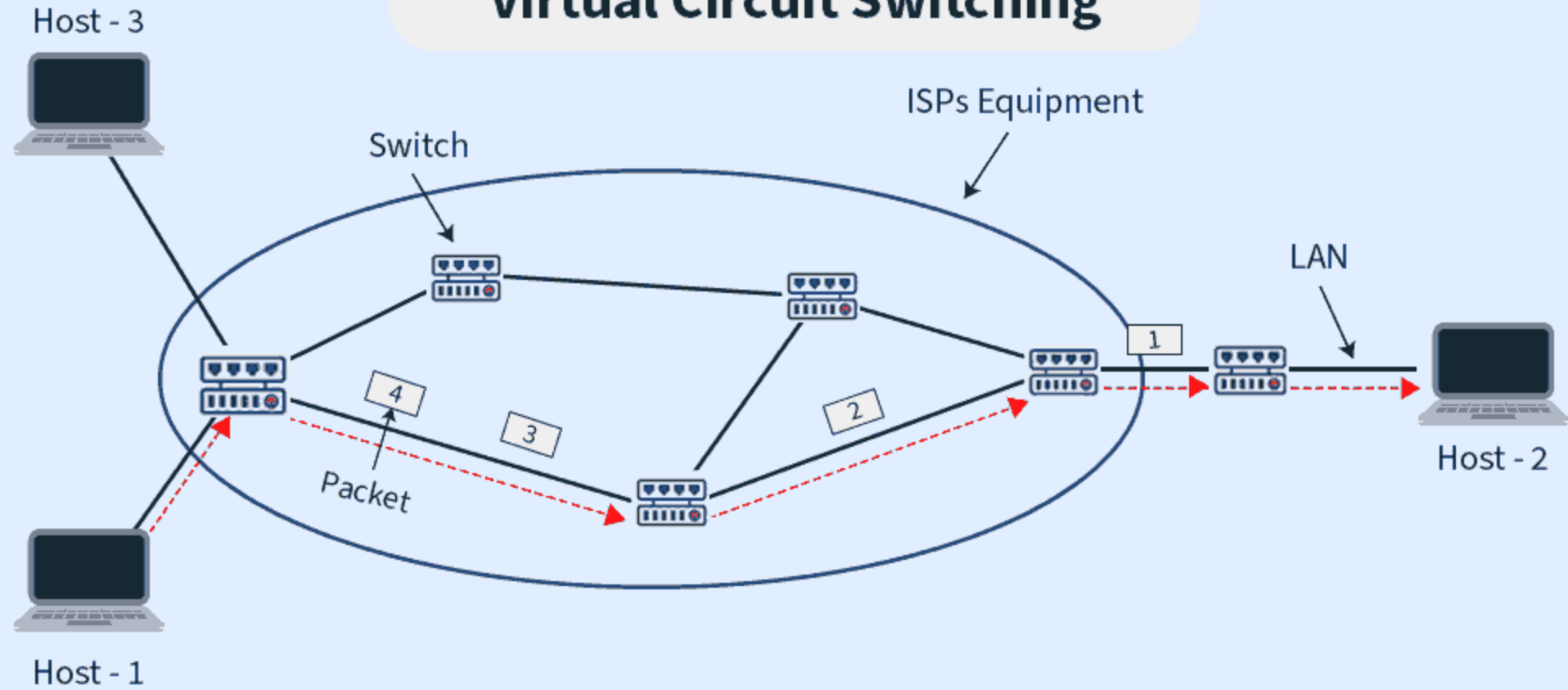
3. Teardown Phase

- Connection is released

Example

- ATM (Asynchronous Transfer Mode)
- Frame Relay
- MPLS (conceptually similar)

Virtual Circuit Switching



Characteristics

- Logical connection exists
- Same path for all packets
- Packets arrive in order
- Lower delay variation

Advantages

- Ordered packet delivery
- Better QoS support
- Less overhead per packet
- Suitable for voice & video

Disadvantages

- Setup delay
- Less flexible than datagram
- Path failure affects communication

VIRTUAL-CIRCUIT NETWORKS

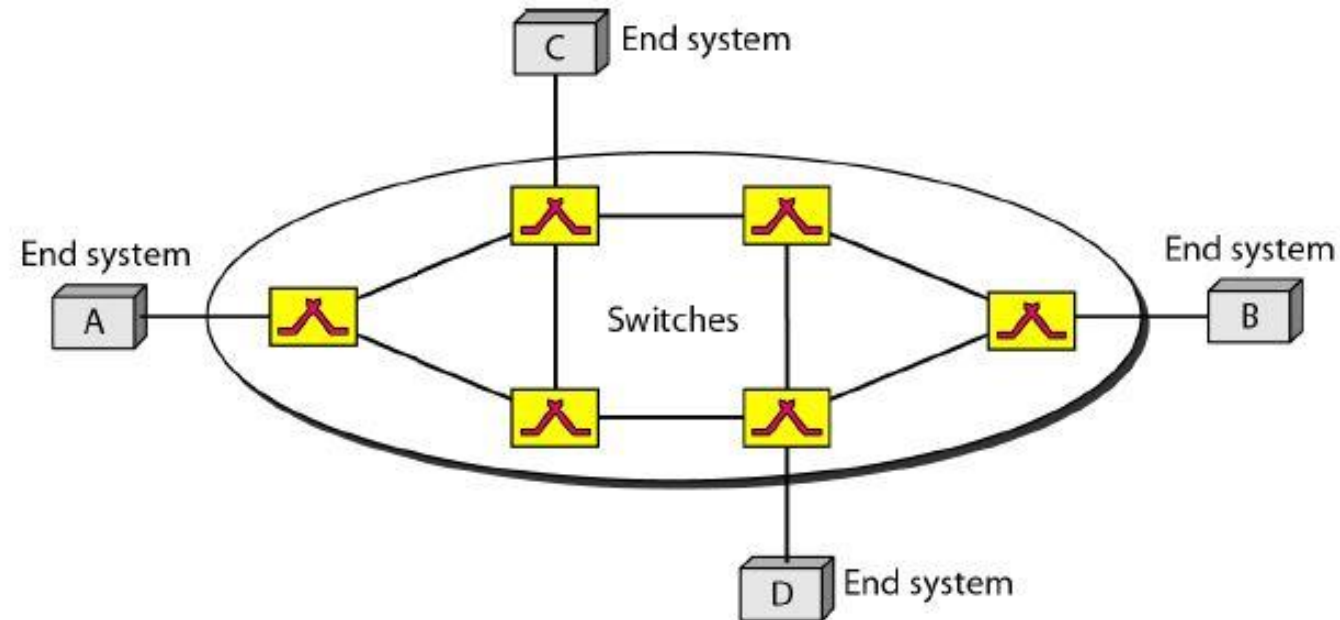
- is a cross between a circuit-switched network and a datagram network
- As in a circuit-switched network, there are setup and teardown phases
- Resources can be allocated during the setup phase, as in a circuit-switched network, or on demand, as in a datagram network.
- As in a datagram network, data are packetized
- each packet carries an address in the header. the address in the header has local jurisdiction. HOW?
 - The answer will be clear when we discuss virtual-circuit identifiers in the next section.

all packets follow the same path established during the connection.

A virtual-circuit network is normally implemented in the data link layer

- circuit-switched network is implemented in the physical layer
- and a datagram network in the network layer.
- But this may change in the future.

VIRTUAL-CIRCUIT NETWORKS



VIRTUAL-CIRCUIT NETWORKS

Addressing

- global Addressing
- Local Addressing

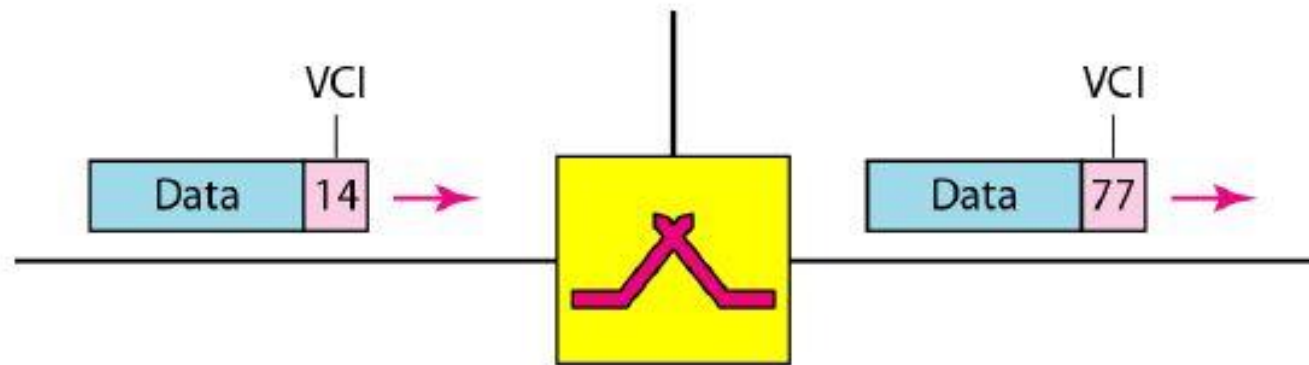
Global Addressing

- A source or a destination needs to have a global address that can be unique and is used only to create a virtual-circuit identifier

Virtual-Circuit Identifier

- The identifier that is actually used for data (VCI)
- VCI does not need to be a large since each switch can use its own unique set of VCIs

VIRTUAL-CIRCUIT Identifier



Three Phases

Setup phase

- the source and destination use their global addresses to help switches make table entries for the connection

data transfer phase

- We discuss first in the next section

Teardown phase

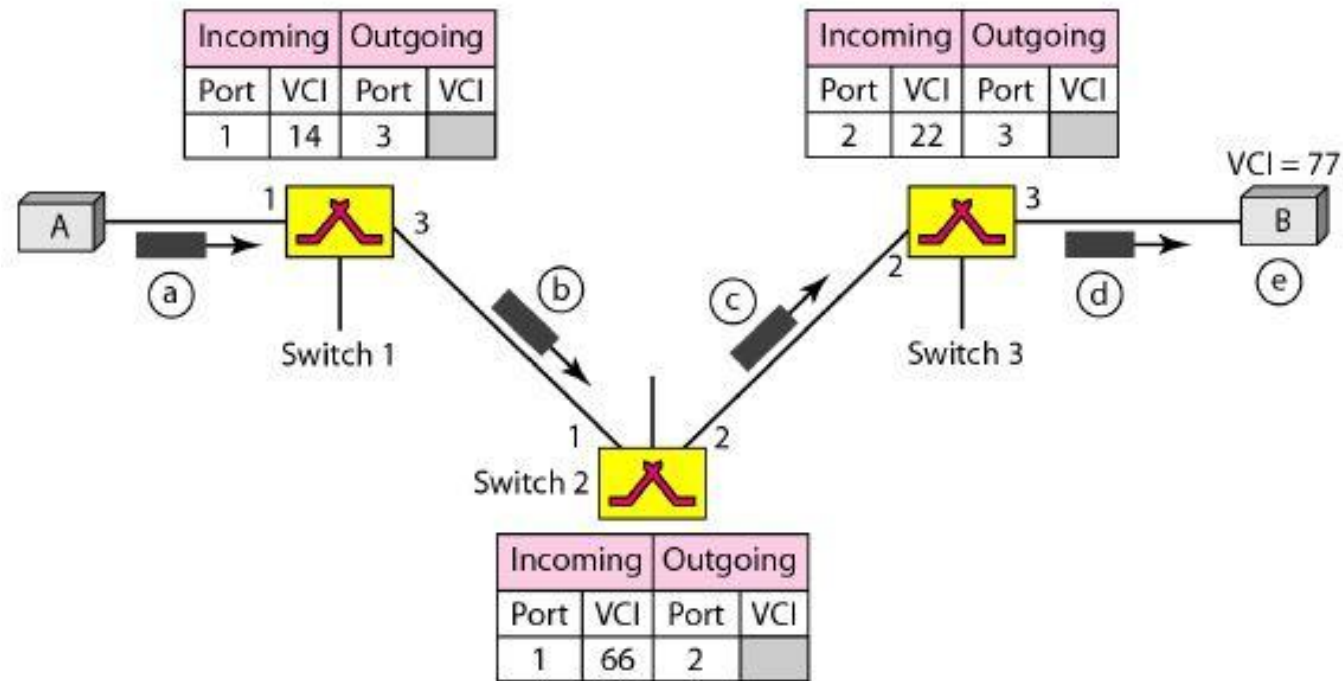
- the source and destination inform the switches to delete the corresponding entry

Setup Phase

setup request

- a) Source A sends a setup frame to switch 1.
- b) Switch 1 receives the setup request frame. It knows that a frame going from A to B goes out through port 3. How? The switch, in the setup phase, acts as a packet switch so through routing table. The switch creates an entry in its table for this virtual circuit, but fill three of the four columns. The switch assigns the incoming port (1) and chooses an available incoming VCI (14) and the outgoing port (3). It does not yet know the outgoing VCI, which will be found during the acknowledgment step. The switch then forwards the frame through port 3 to switch 2.
- c) Switch 2 receives the setup request frame. The same events happen here as at switch 1;
- d) ...
- e) Destination B receives the setup frame, and if it is ready to receive frames from A, it assigns a VCI to the incoming frames that come from A, in this case 77. This VCI lets the destination know that the frames come from A, and not other sources acknowledgment

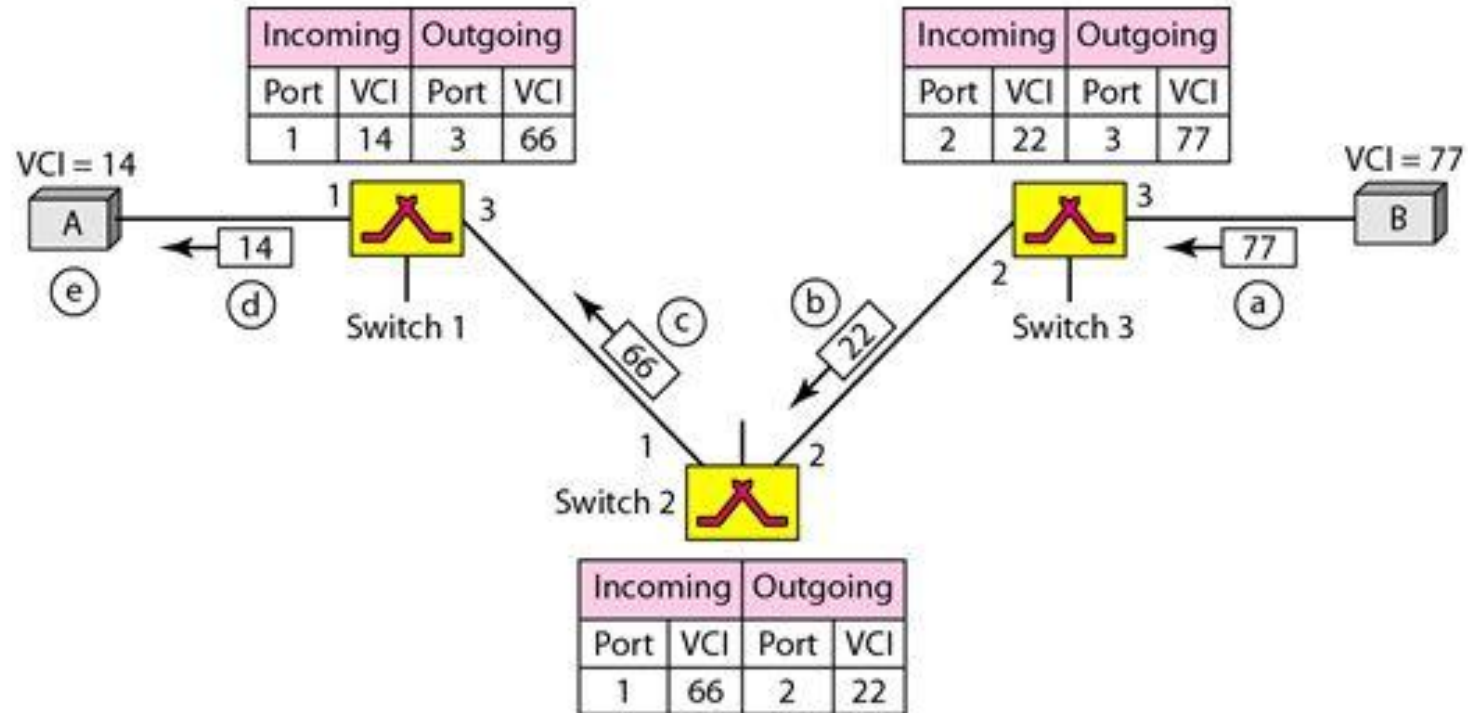
Setup Phase/Request



Setup Phase/Acknowledge

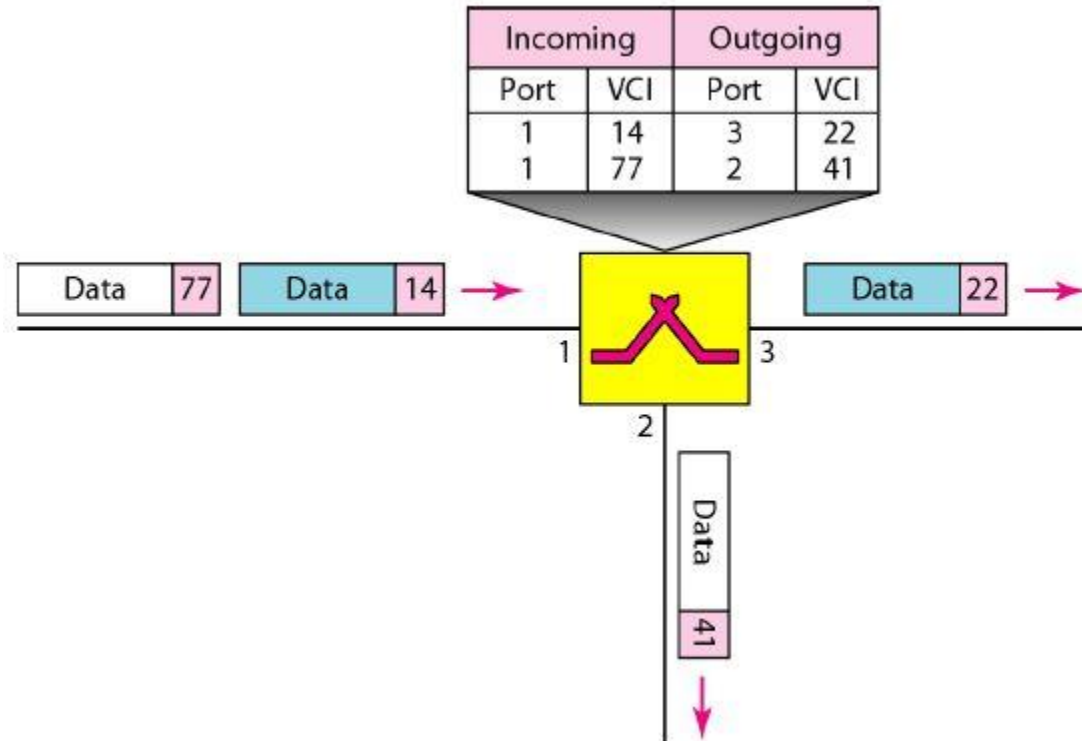
- A. The destination sends an acknowledgment to switch 3. The acknowledgment carries the global source and destination addresses. The frame also carries VCI 77, chosen by the destination as the incoming VCI for frames from A. Switch 3 uses this VCI to complete the outgoing VCI column for this entry. Note that 77 is the incoming VCI for destination B, but the outgoing VCI for switch 3.
- B. Switch 3 sends an acknowledgment to switch 2 that contains its incoming VCI in the table, chosen in the previous step. Switch 2 uses this as the outgoing VCI in the table.
- C. Switch 2 sends an acknowledgment to switch 1 that contains its incoming VCI in the table, chosen in the previous step. Switch 1 uses this as the outgoing VCI in the table.
- D. Finally switch 1 sends an acknowledgment to source A that contains its incoming VCI in the table, chosen in the previous step.
- E. The source uses this as the outgoing VCI for the data frames to be sent to destination B.

Setup Phase/Acknowledge

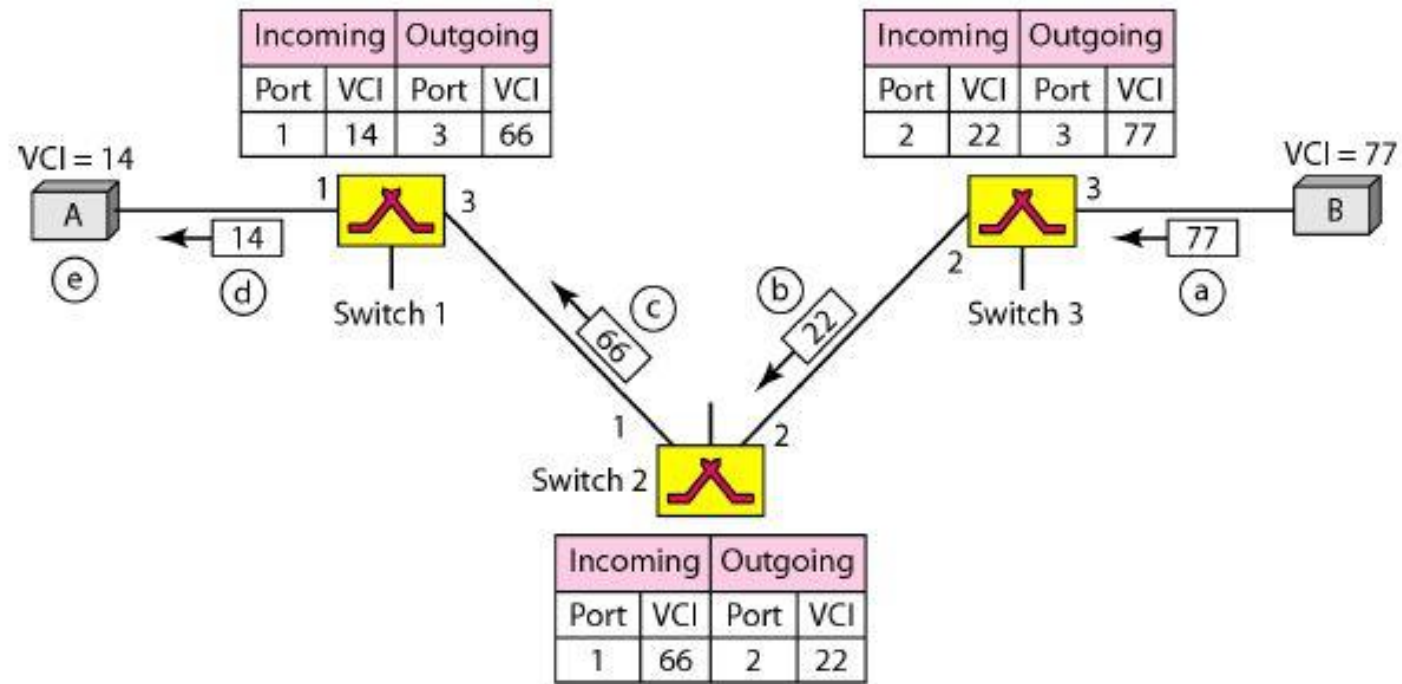


Data Transfer Phase

switch holds four pieces of information for each virtual circuit that is already set up



Data Transfer Phase



Teardown Phase

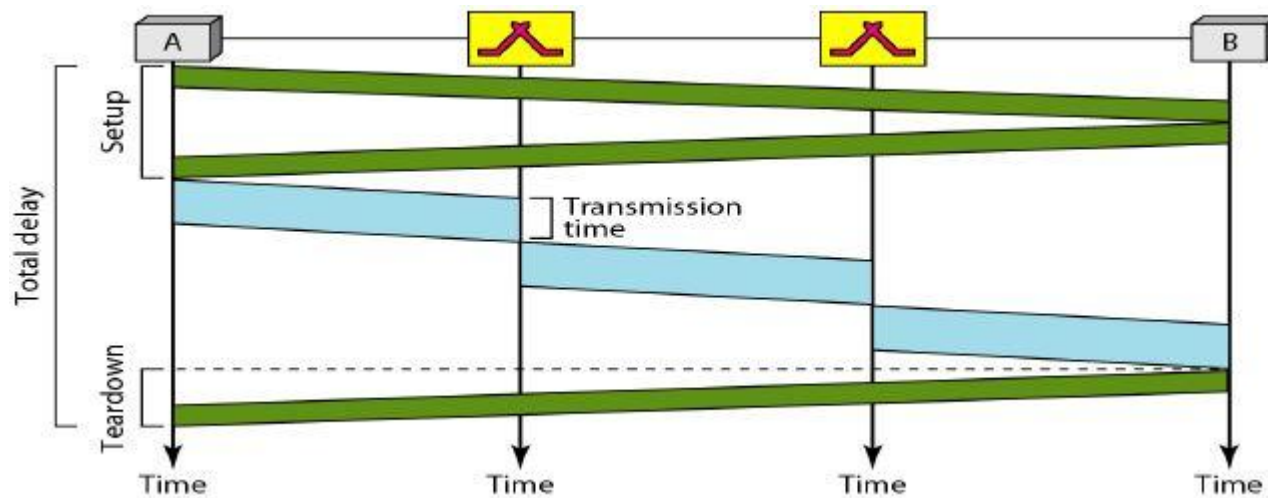
- A. source A, after sending all frames to B, sends a special frame called a *teardown request*
- B. Destination B responds with a teardown confirmation frame
- C. switches delete the corresponding entry from their tables

Delay in Virtual-Circuit Networks

one-time delay for setup (in two direction)

one-time delay for teardown (in one direction)

If resources are allocated during setup phase, there is no wait time for individual packets.



Total delay =
 $3T + 3t + \text{setup delay} + \text{teardown delay}$
ignore the processing time in each switch

Virtual Circuit-Switched Technology in WANs

- Frame Relay and ATM networks
- Implemented in data link layer

Feature	Datagram	Virtual Circuit
Connection Setup	No	Yes
Path	Different for each packet	Same for all packets
Packet Order	Out of order	In order
Delay	Variable	Predictable
Example	Internet (IP)	ATM, Frame Relay

Comparison	Virtual Circuit Switching	Circuit Switching
Path Type	Logical (virtual) path	Physical dedicated path
Data Unit	Packets	Continuous data
Setup	Yes	Yes
Resource Reservation	Partial / logical	Full physical
Packet Order	In order	In order
Bandwidth Usage	Shared	Fully reserved
Example	ATM, Frame Relay	Telephone (PSTN)